

RASSEL A SADAT

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SUMMARY

Curious and detail-oriented Computer Science graduate student with a strong interest in Data Intelligence, Artificial Intelligence, Machine Learning, and data-driven problem solving. Experienced in Python, machine learning, large language model (LLM) API integration, RESTful API development, data preprocessing, backend development, and MongoDB, building practical, reliable, full-stack solutions that transform raw information into structured data for intelligent systems and better decision-making. Quick to learn new technologies, skilled at solving complex problems, and focused on producing accurate, well-organized work. Eager to contribute analytical thinking, attention to detail, and a continuous learning mindset to AI-driven products and data-intensive workflows.

EDUCATION

VIT AP University, Amaravati, Andhra Pradesh	2022 - Present
Integrated M.Tech in Computer Science Engineering Current CGPA: 8.73	
Chinmaya Vidyalaya, Thrissur, Kerala	2020 - 2022
Class XII (CBSE) Class X (CBSE)	

EXPERIENCE

Flipkart GRID 6.0 - Software Development Track	Aug 2024 - Sep 2024
Participant	
- Collaborated in a national-level software engineering competition to solve real-world technical challenges under strict timelines. - Applied data structures, algorithms, and analytical thinking to design scalable and efficient software solutions. - Worked within structured development workflows while emphasizing clean implementation, performance optimization, and effective problem solving.	

PROJECTS

DocuCheck - AI-Powered Document Extraction & Verification System	2026
Technologies: Python, FastAPI, React (Vite), SQLite, Google Gemini API, RESTful APIs	
- Developed a full-stack AI document extraction system using FastAPI and React (Vite) that processes receipts and invoices through the Google Gemini multimodal API. - Designed an independent guardrail validation layer using deterministic arithmetic checks (line-item totals, subtotal and tax reconciliation) to verify AI-extracted data and flag inconsistencies. - Implemented a confidence-based review workflow flagging low-confidence extracted fields for human verification, and built an offline evaluation pipeline that achieved 93.3% field-level extraction accuracy against ground-truth data. - Structured persistent storage with SQLite to separately track raw AI output and human-corrected data for auditability.	
News Aggregator Platform	2024
Technologies: MongoDB, Express.js, React, Node.js	
- Developed a full-stack news aggregation platform that collected information from multiple sources using RESTful APIs. - Built backend services to validate, organize, and categorize incoming data into structured formats for efficient retrieval. - Implemented data filtering and preprocessing techniques to improve the quality and consistency of aggregated information. - Optimized backend processing and API workflows to improve response efficiency and data accessibility.	
Sin1 Hotel Agent - A Survey of Hallucination in Agentic LLM Systems	2026
Technologies: Python, FastAPI, Qwen3 (8B), SQLite, NetworkX, Next.js, RESTful APIs	
- Built a hallucination-resistant hotel booking agent on the Qwen3 (8B) large language model, tested against the Kaggle Hotel Booking dataset (119,390 records), to prevent fabricated policies, fake booking references, and unsupported tool claims. - Designed a five-layer neurosymbolic protection pipeline: Graph-RAG retrieval with 22 NetworkX knowledge graph fact triples, intent-gated SQLite tool execution (ToolGate), dual Qwen3 multi-agent judges, deterministic regex guardrails, and a self-correcting retry loop. - Reduced the Hallucination Risk Score (HRS) by 84.64% (0.5340 → 0.0820) and increased expected safe behavior from 28% to 96% across a structured 25-query benchmark spanning five hallucination categories.	

- Built an interactive Next.js frontend with a live inference console, evaluation dashboard, ablation matrix, and knowledge graph explorer for side-by-side comparison of baseline versus protected model outputs.

SKILLS

Programming Languages: Python, Java, JavaScript, HTML, CSS

Data & AI: Data Collection, Data Cleaning, Data Preprocessing, Data Analysis, Structured Data, Semi-Structured Data, Machine Learning, Random Forest, Model Evaluation (ROC-AUC), Large Language Models (LLM), Google Gemini API, AI Integration, Data Validation

Backend & Web Development: Node.js, Express.js, FastAPI, RESTful APIs, API Integration, React (Vite), Modular Architecture

Databases: MongoDB, SQLite, NoSQL Data Handling

Core Concepts: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Software Development Lifecycle (SDLC)

Tools: Git, GitHub, VS Code, Figma

LANGUAGES

Malayalam - Native | English - Professional | Hindi - Intermediate